

INTERNSHIP PROJECT REPORT

Employee Attrition Prediction Using Machine Learning

Project Type: Machine Learning / Data Analytics

Dataset: HR Employee Attrition Dataset

Prepared from: Submitted Google Colab / Jupyter Notebook

This report documents the complete workflow, results, and conclusions implemented in the project notebook.

1. Executive Summary

This internship project develops and evaluates supervised machine learning models to predict employee attrition. The objective is to identify employees who may be at risk of leaving an organization by learning patterns from employee-related attributes.

The project uses an HR dataset containing **1,470 employee records and 35 original columns**. Data quality checks found no missing values and no duplicate records. Exploratory Data Analysis (EDA), data cleaning, feature preprocessing, stratified train-test splitting, class-imbalance handling, model training, evaluation, model comparison, feature importance analysis, and sample prediction were completed.

Three models were evaluated: Logistic Regression, Decision Tree, and Random Forest. Although Random Forest achieved the highest overall accuracy, Logistic Regression provided the strongest balance for identifying the minority attrition class based on recall, F1-score, and ROC-AUC.

2. Problem Statement

Employee attrition can create recruitment costs, loss of experience, and operational disruption. The project addresses the problem of predicting whether an employee is likely to leave the organization based on available HR and employment characteristics.

3. Project Objectives

- Explore and understand the HR employee dataset.
- Check the dataset for missing values, duplicate records, and class imbalance.
- Perform exploratory data analysis to study relationships between employee characteristics and attrition.
- Prepare numerical and categorical features for machine learning.
- Train and compare multiple supervised classification algorithms.
- Evaluate models using metrics suitable for an imbalanced target variable.
- Identify important predictive features and demonstrate a sample prediction.

4. Dataset Overview

Dataset property	Value
Total employee records	1,470
Original columns	35
Target variable	Attrition
Attrition = No	1,233 (83.88%)
Attrition = Yes	237 (16.12%)
Missing values	None identified
Duplicate records	None identified

5. Data Understanding and Quality Assessment

The dataset structure and descriptive statistics were examined using standard dataframe inspection methods. Numerical and categorical variables were reviewed separately. The data quality checks confirmed that all 1,470 records were complete and that no duplicate rows were present.

The target distribution showed a substantial imbalance: only 16.12% of employees belonged to the attrition class. This made overall accuracy alone insufficient for judging model quality and motivated the use of balanced class weights and additional evaluation metrics.

6. Exploratory Data Analysis

The notebook analyzed the target distribution and explored attrition patterns using visualizations for overtime, job satisfaction, job role, age, and monthly income.

- Attrition distribution was visualized to highlight the imbalance between employees who stayed and those who left.
- Overtime was compared with attrition to investigate workload-related patterns.
- Job satisfaction was examined against attrition.
- Attrition was compared across job roles.
- Age distributions were compared between attrition classes.
- Monthly income distributions were also examined in relation to attrition.

These analyses were used as exploratory investigation; they provide descriptive patterns and do not by themselves establish causal relationships.

7. Data Cleaning and Feature Preparation

Non-informative features were removed before model training. EmployeeNumber was treated as an identifier, while EmployeeCount, Over18, and StandardHours were removed because they do not provide useful variation for prediction. After this step, the feature matrix contained **30 input features**.

The input data was processed using a ColumnTransformer:

- Numerical features were standardized using StandardScaler.
- Categorical features were converted to numerical representations using One-Hot Encoding.
- Preprocessing was incorporated into machine learning pipelines to keep transformations consistent between training and testing.

8. Train-Test Split and Class Imbalance Handling

The dataset was split using an **80:20 train-test ratio** with `random_state=42`. Stratification was applied to preserve the proportion of attrition and non-attrition cases in both subsets.

Because attrition is the minority class, the models used `class_weight='balanced'`. This increases the importance assigned to the minority class during training and helps avoid a model that simply favors the majority class.

9. Machine Learning Models

9.1 Logistic Regression

A Logistic Regression pipeline was trained as the baseline classifier with `max_iter=1000`, balanced class weights, and `random_state=42`.

9.2 Decision Tree

A Decision Tree classifier was trained with max_depth=5, balanced class weights, and random_state=42.

9.3 Random Forest

A Random Forest classifier was trained with 300 estimators, max_depth=10, balanced class weights, and random_state=42.

10. Model Evaluation

The unseen test set contained 294 records, including 247 non-attrition cases and 47 attrition cases. The following metrics were used: Accuracy, Precision, Recall, F1-score, and ROC-AUC.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.752	0.349	0.638	0.451	0.803
Decision Tree	0.786	0.379	0.532	0.442	0.653
Random Forest	0.840	0.500	0.149	0.230	0.797

11. Interpretation of Results

Random Forest achieved the highest overall accuracy of 84.0%. However, its recall for the attrition class was only 14.9%, meaning that it identified relatively few employees who actually left.

Logistic Regression achieved the highest recall at 63.8%, the highest F1-score at 0.451, and the highest ROC-AUC at 0.803. Its overall accuracy was lower than Random Forest, but it detected a much larger proportion of actual attrition cases.

Decision Tree produced intermediate results, with 78.6% accuracy, 53.2% recall, and an F1-score of 0.442.

Therefore, when the business objective is early identification of employees who may leave, Logistic Regression is the most suitable model among those tested. When only overall accuracy is considered, Random Forest appears strongest, but accuracy is misleading for this imbalanced problem.

12. Confusion Matrix and ROC Analysis

Confusion matrices were generated for the trained models to examine correct and incorrect predictions, including true positives, true negatives, false positives, and false negatives.

ROC curves were also generated to compare the models across classification thresholds. Logistic Regression achieved the highest ROC-AUC score of 0.803, followed closely by Random Forest at 0.797.

13. Feature Importance Analysis

Feature importance was extracted from the Random Forest model. The notebook identified the following leading features:

Rank	Feature	Importance
1	MonthlyIncome	0.0752
2	Age	0.0625
3	TotalWorkingYears	0.0543
4	YearsAtCompany	0.0493
5	DailyRate	0.0471

6	DistanceFromHome	0.0429
7	MonthlyRate	0.0404
8	NumCompaniesWorked	0.0402
9	HourlyRate	0.0401
10	YearsWithCurrManager	0.0398
11	StockOptionLevel	0.0328
12	OverTime_No	0.0326
13	OverTime_Yes	0.0314
14	PercentSalaryHike	0.0298
15	YearsInCurrentRole	0.0289

Feature importance represents predictive contribution within this Random Forest model and should not be interpreted as proof of causation.

14. Sample Prediction

A sample employee from the test set was passed to the Logistic Regression model. The model predicted **No Attrition** with an estimated attrition probability of **28.66%**. This demonstrates how the trained pipeline can be used to generate an individual risk prediction.

15. Business Recommendations

- Use the model as an early-warning and decision-support tool to identify employees who may require additional attention.
- Review overtime, compensation-related factors, job satisfaction, and career progression indicators when investigating attrition risk.
- Prioritize follow-up conversations or retention initiatives for employees identified as higher risk rather than making automated employment decisions.
- Evaluate model thresholds according to business costs, especially the trade-off between missing a likely departure and incorrectly flagging an employee.
- Retrain and monitor the model when newer HR data becomes available to detect performance changes over time.

16. Limitations

- The dataset represents the information available in the submitted project and may not capture every factor influencing employee decisions.
- The target class is imbalanced, which affects how accuracy should be interpreted.
- Feature importance and model predictions indicate statistical patterns, not causal explanations.
- Predictions should not be the sole basis for employment, performance, or personnel decisions.

17. Conclusion

This project successfully implemented a complete supervised machine learning workflow for employee attrition prediction. Starting with 1,470 employee records, the work included data quality checks,

exploratory analysis, removal of non-informative variables, preprocessing, stratified train-test splitting, class-imbalance handling, model training, multi-metric evaluation, ROC analysis, feature importance, and sample prediction.

Random Forest produced the highest overall accuracy at 84%, but Logistic Regression was the strongest model for the stated objective of identifying potential attrition because it achieved the highest recall (63.8%), highest F1-score (0.451), and highest ROC-AUC (0.803) among the tested models. The project demonstrates why model selection should be aligned with the business objective rather than based on accuracy alone.

18. Technologies and Libraries Used

Python was used with Pandas for data manipulation, NumPy for numerical operations, Matplotlib and Seaborn for visualization, and Scikit-learn for preprocessing, model training, pipelines, evaluation metrics, ROC analysis, and feature importance.

End of Report